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Life Tables with Concomitant Information

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Summary

A regression model for the analysis of survival data adjusting for concomitant information is developed. The model presented can lead to the log linear exponential model (Glasser [1967]) and the life table regression model of Cox [1972]. In addition, the model described can be used to analyze data from the commonly employed actuarial life table. A discussion of the special case where one is comparing two survival curves is presented. The methods developed are illustrated using data from a clinical trial investigating treatments for lung cancer.

1. Introduction

Life table methods have long been used by medical statisticians and actuaries to analyze survival data. Until recently, however, there has not been a convenient method of carrying out the analysis in such a way as to adjust for concomitant information. A previous attempt at analyzing survival data with concomitant variables required the assumption that the survival times followed an exponential distribution with a parameter which was some function of the concomitant variables (Feigl and Zelen [1965], Zippin and Armitage [1966] and Glasser [1967]). Cox [1972] followed a different approach and presented a life table regression model which brings concomitant information into the product limit life table described by Kaplan and Meier [1958]. In this paper a regression model is introduced which can be used with the commonly employed actuarial life table method in which the period of follow-up is divided into intervals.

2. A General Life Table Regression Model

Suppose that there is associated with the jth individual under observation a vector of concomitant variables $\mathbf{z}_j = (z_{1j}, \cdots, z_{pj})$. The model under consideration will be specified by a hazard function

$$\mu(t; \mathbf{z}_j) = \mu_0(t) \exp(\mathbf{z}_j \boldsymbol{\beta})$$

where $\boldsymbol{\beta}$ is a $p \times 1$ vector of regression parameters associated with the concomitant variables and $\mu_0(t)$ is a function of t which may be interpreted as the hazard function if $\mathbf{z}_j = \mathbf{0}$ or $\boldsymbol{\beta} = \mathbf{0}$. In the development presented in this paper the period of follow-up is divided into k intervals $(x_i, x_{i+1}]$ for $i = 1, 2, \cdots, k$, where $x_1 = 0$. It will further be assumed $\mu_0(t)$ is constant over each interval so that $\mu_0(t) = \mu_i$ for $x_i < t \leq x_{i+1}$. This assumption implies the survival times within an interval are exponentially distributed.

Individuals entering an interval will be classified into two mutually exclusive sets, and

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the presence of an individual in the set will be denoted by an indicator function which takes values of 0 or 1 depending on absence or presence in the set. Let ψ_{ij} be the indicator for whether or not the jth individual is withdrawn alive from the interval $(x_i, x_{i+1}]$ (withdrawal including survival of the whole interval as well as loss to follow-up or withdrawal during the interval); and Δ_{ij} is the indicator for individuals who die in the interval $(x_i, x_{i+1}]$. Let d_i be the number of deaths in the ith interval. The exact time of death or withdrawal alive from the interval $(x_i, x_{i+1}]$ is $(x_i + t_{ij})$ where $(0 < t_{ij} \leq x_{i+1} - x_i)$ and we shall assume it is known for each individual. Using the results from the death process (see Chiang [1958]) and assuming withdrawal is independent of death, we have

$$\Pr\{j\text{th individual dies at } x_i + t_{ij} \mid \text{individual alive and present at } x_i\}$$
$$\propto \mu_i \exp\{\mathbf{z}_j\boldsymbol{\beta}\} \exp(-\mu_i \exp\{\mathbf{z}_j\boldsymbol{\beta}\}t_{ij})$$

and

$$\Pr\{j\text{th individual withdrawn alive at } x_i + t_{ij} \mid \text{individual alive and present at } x_i\}$$
$$\propto \mu_i \exp(-\mu_i \exp\{\mathbf{z}_j\boldsymbol{\beta}\}t_{ij}).$$

The contribution to the likelihood from the jth individual can be found by multiplying over all intervals, the probability of observing what in fact occurred to that individual. This is proportional to

$$\prod_{i=1}^{k} [\mu_i \exp\{\mathbf{z}_j\boldsymbol{\beta}\} \exp(-\mu_i \exp\{\mathbf{z}_j\boldsymbol{\beta}\}t_{ij})]^{\Delta_{ij}}[\exp(-\mu_i \exp\{\mathbf{z}_j\boldsymbol{\beta}\}t_{ij})]^{\Psi_{ij}}.$$

Hence, using the assumed form for the way in which the covariables come into the hazard, we obtain the log likelihood (except for an added constant) for the group of N individuals present initially

$$\begin{aligned} L(\boldsymbol{\beta}, \boldsymbol{\mu}) &= \sum_{j=1}^{N} \sum_{i=1}^{k} \{\Delta_{ij}[\ln \mu_i + \mathbf{z}_j\boldsymbol{\beta} - \mu_i \exp(\mathbf{z}_j\boldsymbol{\beta})t_{ij}] - \Psi_{ij}\mu_i \exp(\mathbf{z}_j\boldsymbol{\beta})t_{ij}\} \\ &= \sum_{i=1}^{k} \left[d_i \ln \mu_i + \sum_{j \in D_i} \mathbf{z}_j\boldsymbol{\beta} - \mu_i \sum_{j \in R_i} \exp(\mathbf{z}_j\boldsymbol{\beta})t_{ij} \right] \end{aligned} \tag{1}$$

where $\boldsymbol{\mu}$ represents the vector of μ_i's, R_i is the set of individuals who enter the interval $(x_i, x_{i+1}]$ and are at risk of dying in the interval, and D_i is the set of individuals dying in $(x_i, x_{i+1}]$. This development does not rely on the outcome in each interval being the result of an independent experiment on a group of individuals. Setting the derivative with respect to μ_i equal to zero yields

$$\hat{\mu}_i(\boldsymbol{\beta}) = \frac{d_i}{\sum_{j \in R_i} \exp(\mathbf{z}_j\boldsymbol{\beta})t_{ij}} \tag{2}$$

which can be substituted into (1) to give

$$L^*(\boldsymbol{\beta}) = \sum_{i=1}^{k} [d_i(\ln d_i - 1) + \sum_{j \in D_i} \mathbf{z}_j\boldsymbol{\beta} - d_i \ln(\sum_{j \in R_i} \exp(\mathbf{z}_j\boldsymbol{\beta})t_{ij})]. \tag{3}$$

Let $\mathbf{U}^*(\boldsymbol{\beta})$ be the vector of the first partial derivatives of $L^*(\boldsymbol{\beta})$ with respect to the elements of the vector $\boldsymbol{\beta}$. Differentiating (3) with respect to $\boldsymbol{\beta}$ yields

$$\mathbf{U}^*(\boldsymbol{\beta}) = \left[\sum_{i=1}^{k} \left[\sum_{j \in D_i} z_{nj} - d_i \frac{\sum_{j \in R_i} z_{nj} \exp(\mathbf{z}_j\boldsymbol{\beta})t_{ij}}{\sum_{j \in R_i} \exp(\mathbf{z}_j\boldsymbol{\beta})t_{ij}} \right] \right]_{p \times 1}. \tag{4}$$

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The maximum likelihood estimate for $\boldsymbol{\beta}$ can be found by setting (4) equal to zero and solving by an interative method. The estimate of $\boldsymbol{\beta}$ can be used in (2) to obtain the maximum likelihood estimates of the μ_i's.

An asymptotic estimate of the variance covariance matrix of all the parameters can be found in the usual way by using the information matrix, which may be derived from (1). It can be shown that the usual estimate of the variance-covariance matrix of $\boldsymbol{\beta}$ is $\mathbf{I}^*(\hat{\boldsymbol{\beta}})^{-1}$ where $\mathbf{I}^*(\boldsymbol{\beta})$ is minus the matrix of the second derivatives of (3) with respect to $\boldsymbol{\beta}$, hence,

$$\mathbf{I}^*(\boldsymbol{\beta}) = \left[\sum_{i=1}^{k} d_i \left\{ \frac{\sum_{j \in R_i} z_{nj} z_{mj} \exp(\mathbf{z}_j \boldsymbol{\beta}) t_{ij}}{\sum_{j \in R_i} \exp(\mathbf{z}_j \boldsymbol{\beta}) t_{ij}} - \frac{(\sum_{j \in R_i} z_{nj} \exp(\mathbf{z}_j \boldsymbol{\beta}) t_{ij})(\sum_{j \in R_i} z_{mj} \exp(\mathbf{z}_j \boldsymbol{\beta}) t_{ij})}{(\sum_{j \in R_i} \exp(\mathbf{z}_j \boldsymbol{\beta}) t_{ij})^2} \right\} \right]_{p \times p} . \quad (5)$$

The simplest special case of the general life table regression model is to consider the period of follow-up as a single interval. This has the effect of assuming the survival times are exponentially distributed with a parameter which is related to covariables by $\mu(\mathbf{z}_j) = \mu \exp(\mathbf{z}_j \boldsymbol{\beta})$. This model was introduced by Glasser [1967] and is considered in further detail by Prentice [1973].

It is possible to bring into this life table regression model concomitant variables which are associated with the intervals. Suppose there are r_i such variables associated with the interval, $(x_i, x_{i+1}]$, represented by the vector $\mathbf{y}_{ji} = (y_{ji1}, \cdots, y_{jir_i})$ for the jth individual. The hazard function for the jth individual in the ith interval then becomes $\mu_i \exp(\mathbf{z}_j \boldsymbol{\beta} + \mathbf{y}_{ji} \boldsymbol{\gamma}_i)$ where $\boldsymbol{\gamma}_i$ is the vector of parameters associated with the $\mathbf{y}_{ji}$ variables. It should be noted, however, the computation involved for this procedure would often be considerable and it would be essential that there be a sufficient number of deaths in each interval to be able to estimate the parameters. In some situations one might bring a new value for the covariable into the interval but assume the regression coefficient was the same for each interval. In this case the hazard would become $\mu_i \exp(\mathbf{z}_j \boldsymbol{\beta} + \mathbf{y}_{ji} \boldsymbol{\gamma})$. A situation where one might want such a procedure is one in which it is necessary to adjust for length of time on treatment. One way this could be done is to include the variable x_i, the beginning of the interval, for those on the treatment and a 0 for the others. The inclusion of similar variables in a product limit life table regression model has been suggested by Cox [1972].

3. *Some Observational Viewpoints for the Period of Follow-up*

When analyzing data using life table regression methods, it is necessary to divide the period of follow-up into intervals. The choice of intervals will influence the degree to which $\mu_0(t)$ is left an arbitrary function of t. Below, two ways of viewing the period of follow-up are discussed: the actuarial and the product limit. In general, the possible biases resulting from an inappropriate model due to a highly variable $\mu_0(t)$ will decrease as the number of intervals increases. This gain should be balanced, however, with the issues of power and ease of computation, both of which should increase as the number of intervals decreases (provided this gives a reasonable representation of the hazard). The form in which the data are available will also influence the choice of intervals; in some situations it is known only that a death occurred during an interval of time, and one cannot divide the period of follow-up into intervals finer than the available data allows.

3.1 *The Actuarial Approach.*

Actuarial life tables are produced by dividing the period of follow-up into intervals and observing the survivorship in each interval. If exact survival information is available one can use the procedure described above directly to obtain estimates of the regression param-

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eters, variances of the parameters, and the maximum of the likelihood. The result produced by this procedure would be influenced to some extent by the choice of intervals, but if the intervals were small so that the probability of surviving the interval was fairly large, this influence would be minor in most practical situations.

In many cases exact survival information is not available; what is known is simply whether or not an individual survived the interval. For those individuals who die or are censored during the interval $(x_i, x_{i+1}]$ one might substitute the expected length of follow-up during the interval. If one assumes the distribution of censored times is uniform, then one would use the time $(1/2)(x_{i+1} - x_i)$ for these individuals. For those individuals who die during the interval, the expected length of follow-up during the interval is

$$-(x_{i+1} - x_i)\left[\frac{1}{\ln p(x_i, \mathbf{z}_j)} + \frac{p(x_i, \mathbf{z}_j)}{1 - p(x_i, \mathbf{z}_j)}\right]$$

where $p(x_i, \mathbf{z}_j)$ is the probability an individual with covariables $\mathbf{z}_j$ will survive the interval $(x_i, x_{i+1}]$. If the probability of surviving an interval is fairly large, one might use $(1/2)(x_{i+1} - x_i)$ and come fairly close to the expected length of follow-up for those who die during the interval (see Table 1). While it is true that this procedure is somewhat crude, if the intervals are selected so that a large proportion of the individuals entering an interval are observed for the entire interval, then it will usually make little difference which convention one adopts. Kalbfleisch and Prentice [1973] suggest a model similar to that discussed in this section when one only has information on survival status during an interval.

3.2 Product-limit Approach.

The general life table regression model can be adapted so the estimation of the regression coefficients is not influenced by the functional form of $\mu_0(t)$ so the procedure is essentially nonparametric. This can be accomplished by using the product limit approach that is described by Kaplan and Meier [1958]. Let the times at which deaths occur be represented by $t_{(1)} < t_{(2)} < \cdots < t_{(h)}$. In the division of the period of follow-up into intervals, let the intervals be chosen so that deaths occur at points at which divisions are made and let the

Table 1

THE EXPECTED VALUE OF THE PROPORTION OF THE INTERVAL LIVED BY THOSE WHO DIE DURING THE INTERVAL, FOR VARIOUS VALUES OF THE PROBABILITY OF SURVIVING THE INTERVAL, ASSUMING EXPONENTIAL SURVIVAL DURING THE INTERVAL

Probability of surviving interval	Expected proportion of interval observed
0.99	0.499
0.95	0.496
0.90	0.491
0.85	0.486
0.80	0.481
0.75	0.476
0.70	0.470

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size of the intervals be arbitrarily small. One might suspect that use of the product limit method will adversely affect the power of the procedure. However, Kalbfleish [1974] gives calculations for the two-sample case which indicates there is little loss of power when exponential survival is appropriate. When the intervals are selected in this way, each individual entering an interval is observed during the entire length of the interval, Δ. A rearrangement of (2) yields

$$\Delta\hat{\mu}_i = \frac{d_i}{\sum_{j \in R_i} \exp(\mathbf{z}_j\boldsymbol{\beta})} \tag{6}$$

and the log likelihood function given in (3) can be written as

$$L(\boldsymbol{\beta}) = \sum_{i=1}^{h} \{ \sum_{j \in D_i} \mathbf{z}_j\boldsymbol{\beta} - d_i \ln (\sum_{j \in R_i} \exp(\mathbf{z}_j\boldsymbol{\beta})) \} + C \tag{7}$$

where C is independent of $\boldsymbol{\beta}$. The asymptotic behavior of the maximum likelihood procedure in this case is not clear because the number of parameters (i.e., the number of μ's) approaches infinity. In addition, the endpoints of the intervals are now stochastic, which is not the case in the actuarial approach. However, the likelihood given in (7) agrees well with that given by Cox [1972] which was based on a conditional argument and that of Kalbfleisch and Prentice [1973] who used a marginal likelihood approach.

The likelihood given in (7) is the same as that given by Cox [1972] and Kalbfleisch and Prentice [1973] if not more than one death occurs at each death time; however, there is some difference when ties in the death times exist. The result in (7) can be obtained using Cox's conditional argument if one adopts the suggestion given by Peto [1972] of slightly modifying the probability that the deaths occur at a particular time conditional on those at risk of dying. In most situations the difference between the two results would be negligible and the likelihood given in (7) would somewhat simplify the computations. It should be noted that, because of the close relationship of this model to that of Cox [1972] and others, the property of rank invariance will hold here as well.

The derivation given in this paper assumed the distributions of survival times was continuous and no allowance was considered for ties in the observation of survival times. In most situations, survival times would be expected to follow some continuous distribution so that ties would only occur because the measurement of time was somewhat crude. Results obtained by using (7) would usually agree quite well with the methods given by Cox [1972] and Kalbfleish and Prentice [1973] when ties exist; however, it is computationally simpler and may be regarded as an approximation to these approaches.

One can obtain an estimate of $\Delta\mu_i$ by substituting into (6) the estimates of $\boldsymbol{\beta}$, obtained by maximizing (7). If the estimates $\hat{\boldsymbol{\beta}} = \mathbf{0}$ or all the covariates are 0, this will estimate the probability of dying at the ith death time. Cox [1972] suggests an iterative procedure for estimating μ_i .

Another choice of intervals which may be used is to divide the period of follow-up at the points at which deaths occur and then assume the hazard is constant over the interval. This procedure is discussed by Breslow [1974].

4. Two Sample Case

Consider the case in which two groups of individuals, Group A and Group B, are being compared. This case can be treated by introducing an indicator variable, z_j for the jth individual where z_j takes the value of 1 if the individual is in Group A and 0 otherwise.

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If we are testing the null hypothesis, $H_0 : \beta = 0$, let us consider as our test statistic $\mathbf{U}^*(\mathbf{0})^T\mathbf{I}^{*-1}(\mathbf{0})\mathbf{U}^*(\mathbf{0})$ which is asymptotically distributed as χ^2 with p degrees of freedom (d.f.) where p is the number of parameters in this case one.[1] From (4) we obtain

$$U^*(0) = \sum_{i=1}^{k} [d_{Ai} - d_i P_{Ai}] \tag{8}$$

where d_{Ai} is the number of deaths occurring in Group A during the ith interval. If T_{Ai} and T_{Bi} are the total length of follow-up in the ith interval for individuals in Groups A and B, respectively, then $P_{Ai} = T_{Ai}/(T_{Ai} + T_{Bi})$. Since P_{Ai} is the proportion of the follow-up time in the ith interval observed for those in Group A, then $d_i P_{Ai}$ is a random variable which estimates, under H_0 , the expected number of deaths occurring in the ith interval in Group A. The statistic under consideration, $U^*(0)$, is the difference between the total number of deaths observed and the total number expected.

The asymptotic variance of $U^*(0)$ can be estimated by

$$I^*(0) = \sum_{i=1}^{k} d_i P_{Ai}(1 - P_{Ai}). \tag{9}$$

Our test statistic becomes

$$\begin{aligned} \chi_1^2 &= \left[\sum_{i=1}^{k} (d_{Ai} - d_i P_{Ai})\right]^2 \Big/ \left[\sum_{i=1}^{k} d_i P_{Ai}(1 - P_{Ai})\right] \\ &= \left[\sum_{i=1}^{k} w_i m_i\right]^2 \Big/ \left[\sum_{i=1}^{k} w_i\right] \end{aligned} \tag{10}$$

where $m_i = \{(T_{Ai} + T_{Bi})/d_i\}\{(d_{Ai}/T_{Ai}) - (d_{Bi}/T_{Bi})\}$ and $w_i = d_i P_{Ai}(1 - P_{Ai})$. The quantity m_i is a standardized difference in the failure rates during the ith interval and w_i is an estimate of Var $(m_i)^{-1}$. The weighted average,

$$\bar{m} = \left[\sum_{i=1}^{k} w_i m_i\right] \Big/ \left[\sum_{i=1}^{k} w_i\right],$$

measures the overall association between the two groups, and is equal to the estimate of β after the first iteration of the Newton-Raphson procedure starting with an initial estimate of zero.

The test described above is reminiscent of the procedure described by Mantel [1966] for comparing the survival for two groups. If the number of deaths in the interval and the number withdrawn alive before the end of the interval are small then the test described above will agree quite closely with Mantel's test. For the product limit case of Section 3.2 where the intervals become small and numerous, the relationship between this procedure and procedures for combining 2×2 tables (see Mantel and Haenszel [1959], Mantel [1963] and Cochran [1954]) has already been noted by Peto and Peto [1972] and Cox [1972]. In the product-limit case the value of $U^*(0)$ given in (8) is the same as that given by Cox [1972], however, $I^*(0)$ given in (9) will be somewhat greater than that given by Cox when ties in the death times exist. Hence, the χ^2 value obtained by the method described above will be smaller than that given by the Cox procedure when ties exist. For the product limit case this test has been shown to be an asymptotically efficient rank invariant test. A permutational argument has also been presented for this test (see Peto and Peto [1972]).

* This arises from the fact that under the null hypothesis, $\mathbf{U}(\mathbf{0})$ is asymptotically multivariate normal with zero mean vector and covariance matrix $\mathbf{I}(\mathbf{0})$.

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The procedure just described will test the hypothesis that the parameter β, which is the same over the entire period of observation, is significantly different from 0. However, the effect of being in Group A might change from interval to interval. An overall test for the differences in survival between Group A and Group B is found by introducing an indicator variable for treatment in each interval. This will lead one to consider

$$\mathbf{U}^*(0) = \begin{bmatrix} d_{A1} - d_1 P_{A1} \\ \vdots \\ d_{Ak} - d_k P_{Ak} \end{bmatrix}_{k\times 1}$$

that is the vector of differences between the observed and the expected number of deaths in each interval. The asymptotic variance-covariance matrix for this vector can be estimated by

$$\mathbf{I}^*(0) = \begin{bmatrix} d_1 P_{A1}(1 - P_{A1}) & & \mathbf{0} \\ & \ddots & \\ \mathbf{0} & & d_k P_{Ak}(1 - P_{Ak}) \end{bmatrix}_{k\times k}.$$

Hence, an overall test for the difference between survival in Group A and survival in Group B, is given by the test statistic

$$\begin{aligned} \chi_k^2 &= \sum_{i=1}^{k} \frac{(d_{Ai} - d_i P_{Ai})^2}{d_i P_{Ai}(1 - P_{Ai})} \\ &= \sum_{i=1}^{k} w_i m_i^2. \end{aligned} \tag{11}$$

An additional test statistic is of some interest, that is the χ^2 on $(k - 1)$ d.f. found by taking the difference between the χ^2 given in (11) and that given in (10). This may also be written as

$$\chi_{(k-1)}^2 = \sum_{i=1}^{k} w_i (m_i - \bar{m})^2$$

which is a test statistic for the homogeneity of the association over all intervals. The procedure described here is analogous to methods used for partitioning χ^2 in the combination of 2×2 tables (see Fleiss [1973]).

In most cases the latter two tests will be quite weak tests of treatment differences. However, if either were significant, one must take them into consideration when drawing conclusions from data. Some other partitions of the overall χ^2 may also be used to give more powerful tests of some specific changes of treatment effect with time. Of particular interest would be linear and quadratic changes in the treatment effect with time.

5. *Example*

Prentice [1973] presents some data collected by the Veterans Administration Lung Cancer Study Group on the survival of 137 advanced lung cancer patients. In the study the patients were randomly assigned to either a standard therapy or a test therapy.

Table 2 presents a comparison of the two treatments set up as in the typical actuarial method. In the present analysis the exact survival times were ignored and the assumption was used that those withdrawn before the end of an interval, either from death or censoring, were observed for half the interval. Because we are not considering the exact survival times

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Table 2

SURVIVAL DATA FOR 137 LUNG CANCER PATIENTS ON STANDARD THERAPY AND TEST THERAPY SUMMARIZED IN AN ACTUARIAL LIFE TABLE (DATA FROM PRENTICE [1973])

FOLLOW-UP INTERVAL IN DAYS	NO. DYING DURING THE INTERVAL $(x_i, x_{i+1}]$		NO. WITHDRAWN ALIVE BEFORE END OF THE INTERVAL $(x_i, x_{i+1}]$		NO. AT RISK IN THE INTERVAL $(x_i, x_{i+1}]$		EXP. PROP. TEST DEATHS	EXP. NO. TEST DEATHS	VARIANCE
	STD.	TEST	STD.	TEST	STD.	TEST			
1-20	15	14	0	0	69	68	0.49796	14.44	7.25
21-40	6	11	1	0	54	54	0.48990	8.33	4.25
41-60	7	10	0	0	47	43	0.46626	7.93	4.23
61-80	2	4	0	0	40	33	0.44286	2.66	1.48
81-100	4	6	2	2	38	29	0.41667	4.17	2.43
1010150	12	5	1	1	32	21	0.41379	7.03	4.12
151-200	7	2	1	0	19	15	0.48276	4.34	2.25
201-250	3	3	0	1	11	13	0.53659	3.22	1.49
251-300	3	1	0	0	8	9	0.56667	2.27	0.98
300+	[5]*	[8]*	0	0	5	8	-------	----	----
TOTAL	59	56						54.38	28.48

*Not included in the total

Table 3

LIFE TABLE REGRESSION ANALYSIS OF LUNG CANCER DATA USING LIKELIHOOD RATIO CHI-SQUARE TESTS

Source	d.f.	Exponential Model $L(\hat{\beta})$	Exponential Model χ^2	Actuarial Model $L(\hat{\beta})$	Actuarial Model χ^2	Product Limit Model $L(\hat{\beta})$	Product Limit Model χ^2
No variables		-751.2212		-743.1878		-586.9063	
Treatment	1	-751.0833	0.28	-743.1847	0.01	-586.9022	0.01
Performance status	1	-726.0955	50.25*	-721.8408	42.69*	-566.0932	41.63*
Treatment, performance status	2	-725.8423	50.76*	-721.5147	43.35*	-565.6446	42.52*
Treatment adjusted for performance status	1		0.51		0.65		0.90
Performance status adjusted for treatment	1		50.48*		43.34*		42.52*
Performance status, cell type	4	-716.9721	68.50*	-712.8611	60.65*	-557.5209	58.77*
Treatment, performance status, cell type	5	-716.4305	69.58*	-712.1185	62.14*	-556.6983	60.42*
Treatment adjusted for performance status, cell type	1		1.08		1.49		1.65
Cell type adjusted for treatment, performance status	3		18.82*		18.39*		17.89*
Treatment, performance status, cell type, treatment-cell type interaction	8	-713.7752	74.89*	-708.5883	69.20*	-553.4544	66.90*
Treatment-cell type interaction adjusted for treatment, performance status, cell type	3		5.31		7.06		6.49

*$p < 0.01$

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at present, it is not of interest to consider further the last interval, since all individuals in this interval die, hence the expected number of deaths equals the observed. A test of whether the test therapy had an effect different from the standard can be made using (10). This gives a χ^2 on one d.f. of 0.09, indicating no difference. An overall test of any treatment differences in survival between the treatment groups over time is given by the χ^2 on nine d.f. of 10.44, which is not significant. A test of whether or not the treatment effect changes from interval to interval is given by the χ^2 on eight d.f. of $10.44 - 0.09 = 10.35$, which is again not significant.

These data included lung cancer patients with different cell types (squamous, small, adeno and large) and it was of interest to adjust for this as well as to see if there was an interaction of treatment with cell type. In addition, performance status was adjusted for, a variable which took the values 10, 20, $\cdots$, 90 (10, 20, 30—completely hospitalized; 40, 50, 60—partial confinement to hospital, and 70, 80, 90—able to care for self). For this further analysis exact survival time information was used. Table 3 gives a comparison of the results obtained by using exponential survival (which is essentially the analysis done by Prentice [1973]) and an actuarial approach where the period of follow-up was divided into ten intervals at the points 20, 40, 60, 80, 100, 150, 200, 250 and 300 days of follow-up.

Table 4

REGRESSION COEFFICIENTS AND THEIR STANDARD DEVIATIONS FROM THE LIFE TABLE REGRESSION ANALYSIS OF LUNG CANCER PATIENTS WITH AND WITHOUT INTERACTION TERMS INCLUDED

	Exponential		Actuarial (10 intervals)		Product Limit	
	$\hat{\beta}_i$	$sd(\hat{\beta}_i)$	$\hat{\beta}_i$	$sd(\hat{\beta}_i)$	$\hat{\beta}_i$	$sd(\hat{\beta}_i)$
WITHOUT INTERACTIONS						
Performance status	-0.0297	0.0048	-0.0308	0.0051	-0.0311	0.0052
Treatment	0.2012	0.1932	0.2436	0.2000	0.2573	0.2006
Small vs squamous	0.7936	0.2545	0.8472	0.2668	0.8196	0.2688
Adeno vs squamous	1.0819	0.2698	1.1645	0.2939	1.1477	0.2949
Large vs squamous	0.3696	0.2719	0.3908	0.2793	0.3930	0.2822
WITH INTERACTIONS						
Performance status	-0.0293	0.0050	-0.0312	0.0053	-0.0315	0.0054
Treatment	-0.2908	0.3713	-0.3641	0.3864	-0.3355	0.3949
Small vs squamous	0.3366	0.3358	0.3284	0.3365	0.3290	0.3377
Adeno vs squamous	1.0881	0.4396	1.2100	0.4500	1.1966	0.4507
Large vs squamous	-0.0475	0.3907	-0.0674	0.3919	-0.0533	0.3936
Small x treatment	0.9428	0.4830	1.1977	0.5139	1.1416	0.5172
Adeno x treatment	0.0405	0.5733	0.1282	0.5855	0.1247	0.5894
Large x treatment	0.7283	0.5420	0.8501	0.5552	0.8511	0.5657

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The tests given are likelihood ratio chi-square tests.[1] One can test for the improved fit obtained by dividing the period of follow-up into intervals by comparing the maximum of the log likelihoods for the actuarial model and the simple exponential model. This will yield a likelihood ratio chi-square on nine d.f. of 16.07 if no variables are included and 10.37 if all the variables are included. While neither is significant, the value of 16.07 is approaching significance ($p < 0.1$) so that one might make a case for using the actuarial approach here. Also given are the results obtained by using the product limit approach. The comparison of these maximum log likelihoods with those from either the exponential model or the actuarial approach is not possible.

Table 4 gives the regression coefficients obtained from fitting the exponential model, the actuarial life table model, and the product limit life table model. In order to avoid redundancy, cell type was coded by three 1, 0 variables indicating the individual did or did not have lung cancer of the cell type—small, adeno or large. This, in effect, compares each cell type to squamous. The table gives the results with and without the treatment and cell type interaction terms included. This analysis shows that higher performance status is associated with better survival and the hazard varies with cell type (squamous had the best prognosis followed by large, small then adeno). The data gave no strong evidence of treatment differences.

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Tables de Survie et Informations Concomitantes

Résumé

On développe un modèle de régression pour l'analyse de données de survie en tenant compte d'informations concomitantes. Le modèle présenté peut conduire au modèle exponentiel Log-linéaire de Glasser [1967] ou au modèle de régression de table de survie de Cox [1972]. Le modèle décrit peut être utilisé pour analyser les données de table de survie actuarielles utilisées habituellement. On présente une discussion du cas spécial où l'on compare 2 courbes de survie. Les méthodes développées sont illustrées par des données provenant d'un essai clinique de traitement du cancer du poumon.

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[1] The likelihood ratio chi-square test is based on the fact that if Ω is a parameter space and ω is a subspace of Ω to which p constraints have been applied, and if $\hat{\Omega}$ and $\hat{\omega}$ are maximum likelihood estimates of Ω and ω then $-2\ln\,[l(\hat{\omega})/l(\hat{\Omega})]$ asymptotically has a chi-square distribution with p d.f. where $l(\cdot)$ is the likelihood.

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